

PokeMart

A Commerce-Style Pokemon Search

Built with **Next.js (App Router)** + **Coveo Headless React (SSR)**

Repo: github.com/zikorao/CoveoProject

The Brief

Core requirements

- Facet search results by **Pokemon Type**
- Facet search results by **Pokemon Generation**
- Show each Pokemon's **picture** in the results

Stretch goals

- Server-side rendering, analytics, type-ahead, detail pages
- Make it look like a real commerce storefront

Solution at a Glance

- **Storefront UI:** search bar, type/generation facets, product-card grid, pagination, result count
- **Clean data:** ~1025 Pokemon pushed from PokeAPI into a Coveo Push source
- **Type-ahead:** ML Query Suggestions (preloaded) + live instant results
- **Detail page:** per-Pokemon view fetched from the Coveo Search API
- **SSR + analytics:** server fetch, client hydration, per-visitor context

Architecture

```
PokeAPI --(push_pokemon.py)--> Coveo Push API --> Coveo Index
                                     |
                                     Search API
                                     |
Usage Analytics --> QS model

Next.js App Router:
  page.tsx (server, fetchStaticState)
    -> SearchProvider (hydrateStaticState)
      -> SearchInterface (SearchBox / Facets / ResultList / Pager)
  pokemon/[name]/page.tsx (server, single-doc REST)
  middleware.ts -> visitor cookie
```

Tech Stack

- **Next.js 14 App Router, React 18, TypeScript**
- **@coveo/headless-react 2.9.16** (`/ssr` entrypoint)
- **Python 3** (stdlib only) for ingestion + analytics simulation
- **PokeAPI** as the upstream data source

Data Pipeline (Push, not Crawl)

```
scripts/push_pokemon.py
```

- One clean document per Pokemon: `title` , `type` (multi), `generation` , `picture`
- Push API **file-container** flow -> batch ingestion (~1025 docs)
- Credentials from env vars (no keys on disk); idempotent

Why push over crawl? Structured, reliable fields - no HTML noise in facets.

Coveo Configuration

- Engine reads `organizationId` + `accessToken` from env vars
- `analytics: { enabled: true }`
- **Constant query** isolates the clean source:
`cq = @source=="push API solution"`
- Controllers: `searchBox, resultList ( picture/type/generation ), typeFacet, generationFacet, querySummary, pager`
- QS model associated with the **default** pipeline

Search Experience

- Faceted filtering with **official Pokemon type colors** + counts
- Product-card grid: artwork, name, **generation badge**, **type chips**
- Query summary ("Showing 1-10 of N") and pagination
- Fully responsive storefront layout

Type-Ahead: Two Mechanisms

1. ML Query Suggestions (preloaded on focus)

- From the trained QS model; richer as traffic grows

2. Instant Results (live, as-you-type)

- Coveo matches whole words -> we use **wildcard prefix** (`char` -> `char*`)
- Debounced + cancellable (`AbortController`)
- Toggle via `INSTANT_RESULTS_ENABLED`

Together: works today (instant) + improves over time (ML).

Training the QS Model

```
scripts/simulate_searches.py
```

- Logs ~3075 Usage Analytics **search events** (one per Pokemon name)
- Gives the model candidate queries to learn from
- Model **rebuilds daily** to incorporate the data

Cold-start reality: a new catalog has no traffic, so the model starts empty ("Model is empty and won't return responses").

SSR + Analytics

- **Hydration pattern:** `fetchStaticState` (server) -> `hydrateStaticState` (client)
 - Better first paint + SEO, no empty-result flash
- **NavigatorContext + middleware:** forwards referrer, UA, IP, stable visitor id
 - Analytics events correlate across SSR and client interactions

Security Hardening

- Hard-coded token -> `NEXT_PUBLIC_COVE0_*` **environment variables**
- `.env.local` gitignored; `.env.example` committed
- Exposed token **rotated** and **scrubbed from git history**
(`git filter-repo` + force-push)
- Push API key never stored in code

Challenges & Fixes (1/2)

Challenge	Fix
npm ERESOLVE (React 18 vs 19)	Isolated package.json, pinned React 18
UTF-8 build error	Rewrote files in ASCII
Coveo <code>fetch failed</code>	Corrected organization id
Missing navigator context	Added middleware + NavigatorContext
PokeAPI 403	Added User-Agent header

Challenges & Fixes (2/2)

Challenge	Fix
Polluted facet values (mixed sources)	Constant query to clean push source
QS empty (cold-start)	Simulated analytics traffic + daily rebuild
Instant results 0 on partial input	Enabled wildcards + prefix query
Secret in public repo	Env vars + rotation + history scrub
RGA missing in SSR 2.9.16	Documented upgrade / client-only paths

Key Trade-offs

- **Push API** over web crawl -> clean structured data
- **Constant query** over deleting the old source -> non-destructive, reversible
- **SSR + hydration** over pure CSR -> first paint + SEO
- **Instant results + ML QS** over QS-only -> beat the cold-start
- **Direct REST** on detail page over a second engine -> simpler

Roadmap

- Flip `@type` to "Include in results" -> inline type chips everywhere
- Grow search traffic -> richer ML prefix suggestions
- **Relevance Generative Answering (RGA):**
 - Confirm license -> enable on pipeline -> enrich content -> add UI (headless upgrade or client-only `buildGeneratedAnswer`)

Demo

- **Catalog:** facets, product grid, pagination
- **Type-ahead:** focus -> suggestions; type `char` -> instant results
- **Detail page:** click a card -> Pokemon detail

`npm run dev` -> <http://localhost:3000>

Thank You

PokeMart - Coveo Search Solution

github.com/zikorao/CoveoProject

Questions?